

UCab – Full Stack Cab Booking System

Final Report

1. Introduction

Project Title: UCab – Full Stack Cab Booking System Using MERN Stack

Team Members

Name	Role
Aishwarya Varshini Grandhi	Frontend Development, Backend Development, Database Design, API Development, Testing, Documentation

2. Project Overview

Purpose

UCab is a full-stack web application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The main objective of the project is to provide users with a simple, efficient, and secure platform for booking cab rides online. The application enables users to register, log in, book rides, estimate fares, and manage their booking history through a responsive and user-friendly interface.

The project demonstrates the implementation of full-stack web development concepts, including RESTful APIs, authentication, database management, and responsive web design.

Features

- User Registration and Login
- Secure Authentication
- Book Cab Online
- Pickup and Drop Location Selection
- Fare Estimation
- Display Available Cabs
- Booking Confirmation
- Ride History
- Responsive User Interface

- User Profile Management
- MongoDB Database Integration
- RESTful API Communication

3. Architecture

Frontend Architecture

The frontend of UCab is developed using React.js. The application follows a component-based architecture, where each page and UI element is implemented as a reusable component. React Router DOM is used for navigation, while Axios handles communication with the backend APIs. Bootstrap is used to create a responsive and modern user interface.

Technologies Used

- React.js
- React Router DOM
- Bootstrap
- Axios
- HTML5
- CSS3
- JavaScript (ES6)

Backend Architecture

The backend is developed using Node.js and Express.js. Express is used to create RESTful APIs that process user requests, manage bookings, and interact with the MongoDB database. The backend also handles authentication and request validation.

Backend Modules

- Authentication
- User Management
- Cab Booking
- Ride Management
- Fare Calculation

- Booking History

Database Architecture

MongoDB is used as the NoSQL database for storing application data. Mongoose provides schema-based interaction between the application and the database.

Collections

- Users
- Bookings
- Rides
- Payments (Optional/Future Enhancement)

Relationships are maintained using Object IDs where necessary.

4. Setup Instructions

Prerequisites

Before running the project, install the following software:

- Node.js (Version 16 or above)
- npm (Node Package Manager)
- MongoDB Community Server or MongoDB Atlas
- Git
- Visual Studio Code
- Postman

Installation

Step 1: Clone the repository.

```
git clone <repository-url>
```

Step 2: Navigate to the project directory.

```
cd UCab
```

Step 3: Install backend dependencies.

```
cd backend
```

```
npm install
```

Step 4: Install frontend dependencies.

```
npm create vite@latest frontend -- --template react
```

```
cd frontend
```

```
npm install
```

Step 5: Configure the environment variables in the .env file.

Example:

- MongoDB Connection String
- JWT Secret Key
- Server Port

5. Folder Structure

Frontend (Client)

frontend

```
|
|
├── public
|
├── src
|   ├── Components
|   ├── Pages
|   ├── Services
|   ├── Assets
|   ├── App.js
|   └── index.js
|
├── package.json
└── package-lock.json
```

Description

- Components – Reusable UI Components

- Pages – Application Pages
- Services – API Calls
- Assets – Images, Icons, CSS Files

Backend (Server)

backend

```
|
|
|— Config
|— Controllers
|— Middleware
|— Models
|— Routes
|— server.js
|— package.json
└— package-lock.json
```

Description

- Config – Database Configuration
- Controllers – Business Logic
- Middleware – Authentication and Validation
- Models – MongoDB Schemas
- Routes – API Endpoints

6. Running the Application

Backend

Open terminal:

```
cd backend
```

```
npm start
```

Backend runs on: <http://localhost:5000>

Frontend

Open another terminal:

```
cd frontend
```

```
npm start
```

Frontend runs on: <http://localhost:3000>

7. API Documentation

The backend exposes RESTful APIs for handling user authentication, bookings, and ride management.

Authentication APIs

Method	Endpoint	Description
POST	/api/register	Register a new user
POST	/api/login	User login

Booking APIs

Method	Endpoint	Description
GET	/api/bookings	Get all bookings
POST	/api/bookings	Create a booking
GET	/api/bookings/:id	Get booking details
PUT	/api/bookings/:id	Update booking
DELETE	/api/bookings/:id	Cancel booking

User APIs

Method	Endpoint	Description
GET	/api/users	Get user profile

Method	Endpoint	Description
PUT	/api/users/:id	Update profile

Example JSON Response

```
{  
  "success": true,  
  "message": "Booking Created Successfully",  
  "bookingId": "123456"  
}
```

8. Authentication

UCab implements secure user authentication using JSON Web Tokens (JWT).

Authentication Process:

- User registers using email and password.
- Passwords are encrypted before storing in MongoDB.
- During login, credentials are verified.
- A JWT token is generated upon successful authentication.
- The token is used to access protected routes.
- Unauthorized users cannot access restricted resources.

Security Features:

- JWT Authentication
- Password Encryption
- Protected API Routes
- Input Validation

9. User Interface

The user interface is designed to provide a smooth and responsive booking experience.

Major Screens:

- Home Page
- Login Page
- Registration Page
- User Dashboard
- Cab Booking Page
- Booking Confirmation Page
- Booking History
- User Profile

10. Testing

The application was tested to ensure proper functionality of all modules.

Testing Performed:

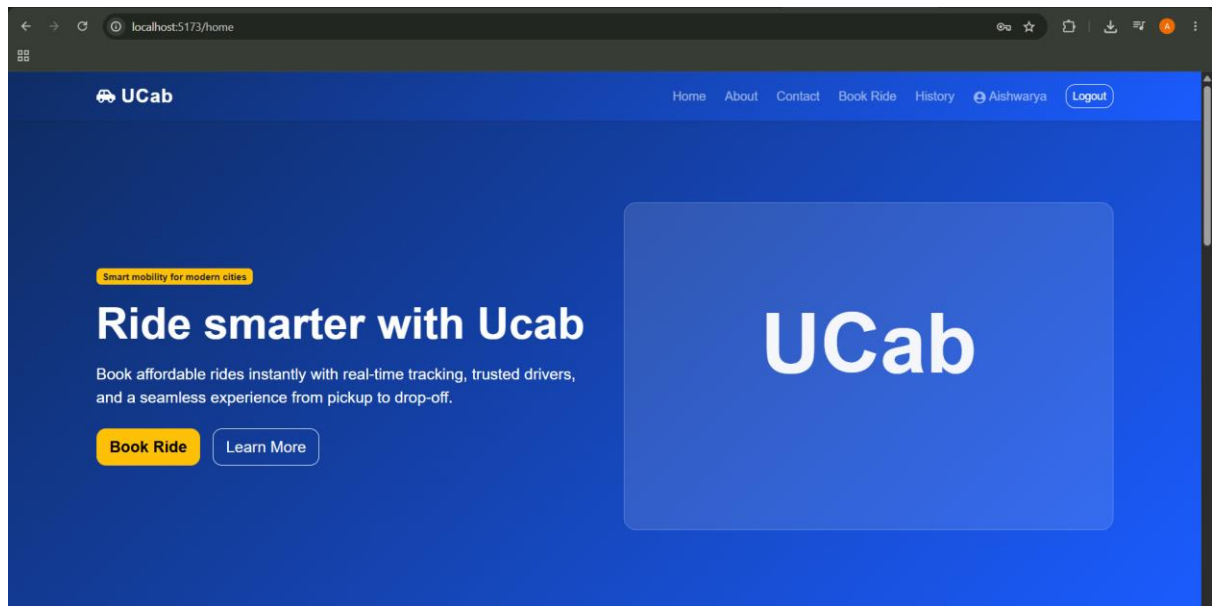
- User Registration
- User Login
- Authentication
- CRUD Operations
- Booking Process
- Database Connectivity
- API Testing using Postman
- Responsive UI Testing
- Form Validation

Testing Tools:

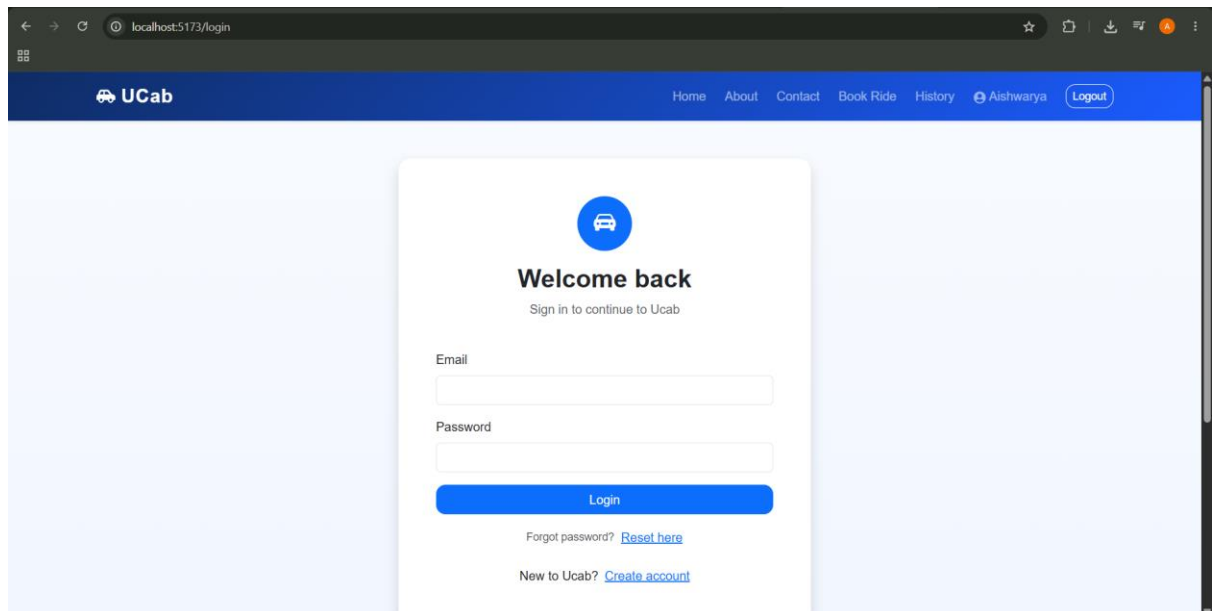
- ThunderClient
- Chrome Developer Tools
- MongoDB Compass

11. Screenshots / Demo

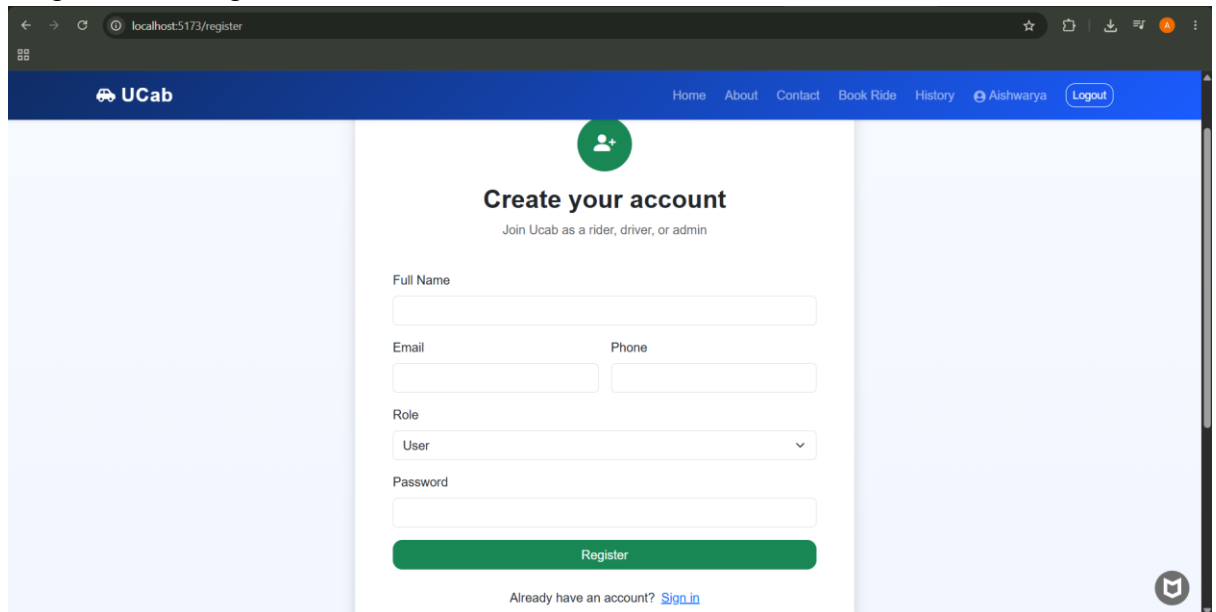
- Home Page



- Login Page

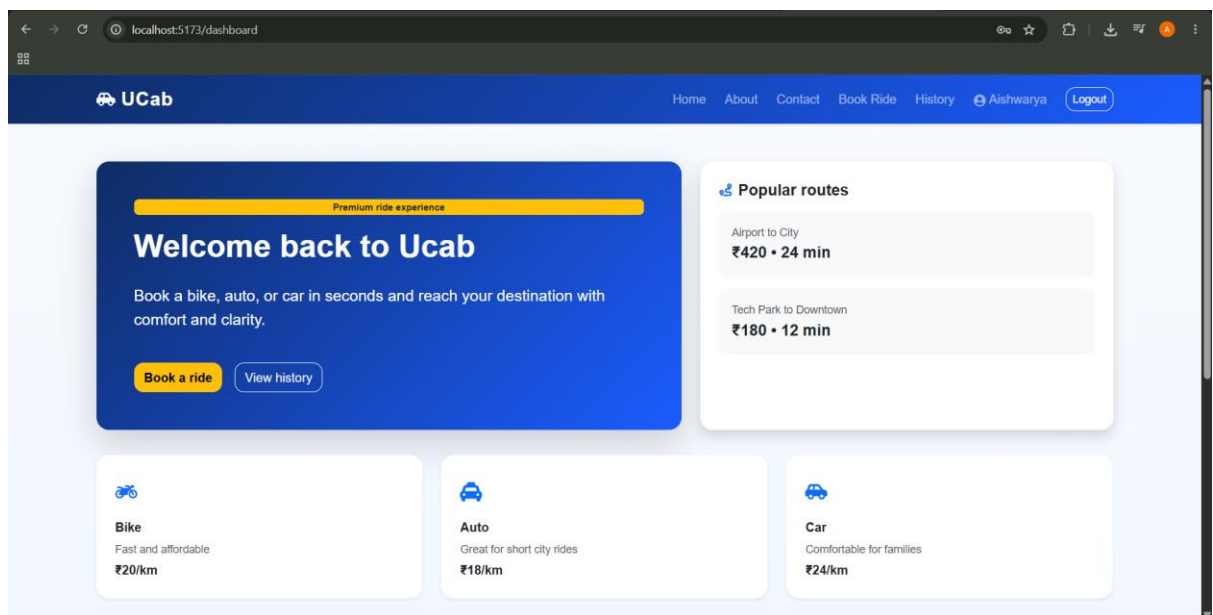


- Registration Page



The screenshot shows the registration page of the Ucab application. The browser address bar displays 'localhost:5173/register'. The page features a dark blue header with the Ucab logo and navigation links: Home, About, Contact, Book Ride, History, Aishwarya, and a Logout button. The main content area is white and centered, with a green circular icon containing a white person silhouette at the top. Below this is the heading 'Create your account' and the subtext 'Join Ucab as a rider, driver, or admin'. The registration form includes input fields for 'Full Name', 'Email', and 'Phone', a 'Role' dropdown menu set to 'User', and a 'Password' field. A green 'Register' button is positioned below the form. At the bottom, a link says 'Already have an account? [Sign in](#)'. A small circular icon with a white 'M' is visible in the bottom right corner.

- User Dashboard



The screenshot displays the user dashboard of the Ucab application. The browser address bar shows 'localhost:5173/dashboard'. The header is identical to the registration page. The dashboard features a large blue welcome banner with the text 'Welcome back to Ucab' and 'Book a bike, auto, or car in seconds and reach your destination with comfort and clarity.' Below the banner are two buttons: 'Book a ride' (yellow) and 'View history' (white). To the right, a 'Popular routes' section lists two options: 'Airport to City' (₹420 • 24 min) and 'Tech Park to Downtown' (₹180 • 12 min). At the bottom, three vehicle options are presented in white cards: 'Bike' (Fast and affordable, ₹20/km), 'Auto' (Great for short city rides, ₹18/km), and 'Car' (Comfortable for families, ₹24/km).

- Cab Booking

The screenshot shows the UCab website interface. The header includes the UCab logo and navigation links: Home, About, Contact, Book Ride, History, Aishwarya, and Logout. The main content area is divided into two columns. The left column, titled 'Book your ride', contains a form for booking a ride. It includes a 'Pickup location' field with 'Cyber towers, Hitech city' and a 'Drop location' field with 'AMB mall'. Below these are three vehicle options: 'Bike' (Fast and budget-friendly), 'Auto' (Great for short city trips), and 'Car' (Comfort for families and business rides). The 'Auto' option is selected. The 'Estimated fare' is shown as ₹305 for a distance of 18 km and ETA of 4 min. A 'Confirm ride' button is at the bottom. The right column, titled 'Ride summary', shows a placeholder for ride details, assigned driver, and payment options.

Book your ride

Choose your ride type, enter your route, and get a live estimate before you confirm.

Pickup location: Cyber towers, Hitech city

Drop location: AMB mall

Choose vehicle:

- Bike**: Fast and budget-friendly
- Auto**: Great for short city trips
- Car**: Comfort for families and business rides

Estimated fare: ₹305
Distance: 18 km • ETA: 4 min

Confirm ride

Ride summary

Your ride details, assigned driver, and payment options will appear here after booking.

- Booking Confirmation

The screenshot shows the UCab website interface after a successful booking. A green notification banner at the top right says 'Ride booked successfully'. The 'Book your ride' form on the left now shows the pickup location as 'Ameerpet metro station' and the drop location as 'Bus station, Mehdiapatnam'. The 'Bike' option is selected. The 'Estimated fare' is ₹227 for a distance of 16 km and ETA of 3 min. The 'Confirm ride' button is still present. The 'Ride summary' section on the right now displays the route 'Ameerpet metro station → Bus station, Mehdiapatnam', the vehicle 'Bike', and the assigned driver 'Kiran' (Bajaj Pulsar, 4.7 rating, Phone: +91 98765 67890). The fare is ₹227 for a distance of 16 km. The vehicle requested is 'Bike'. A 'Pay now' button is at the bottom.

Book your ride

Choose your ride type, enter your route, and get a live estimate before you confirm.

Pickup location: Ameerpet metro station

Drop location: Bus station, Mehdiapatnam

Choose vehicle:

- Bike**: Fast and budget-friendly
- Auto**: Great for short city trips
- Car**: Comfort for families and business rides

Estimated fare: ₹227
Distance: 16 km • ETA: 3 min

Confirm ride

Ride summary

Route: Ameerpet metro station → Bus station, Mehdiapatnam

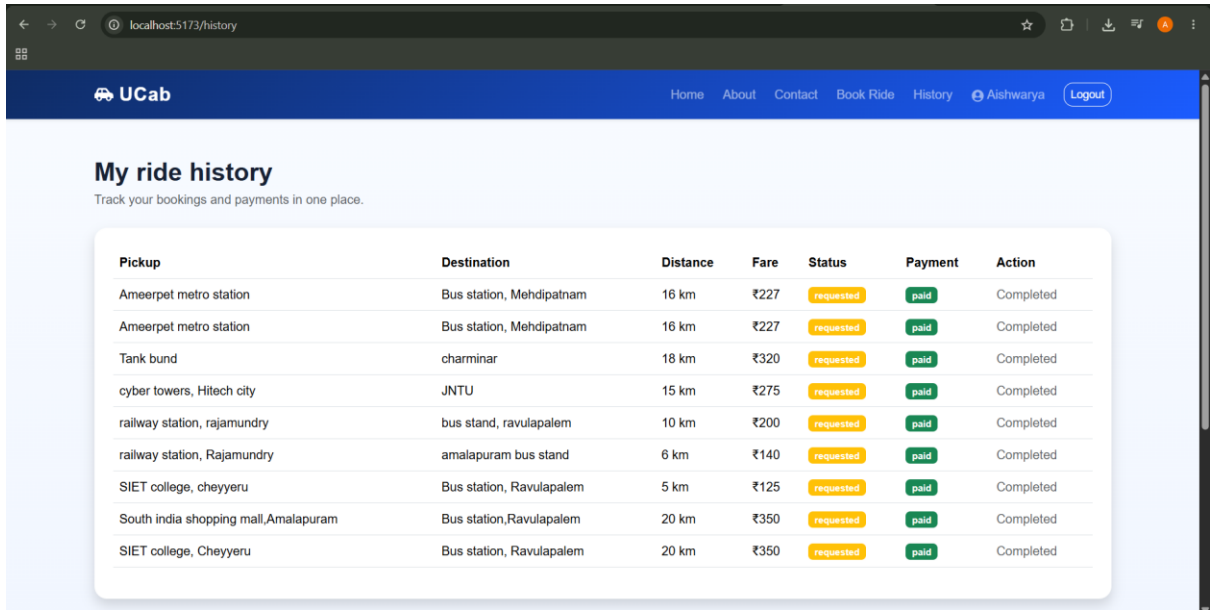
Vehicle: Bike

Assigned driver: **Kiran** (Bajaj Pulsar, 4.7 rating, Phone: +91 98765 67890)

Fare: ₹227
Distance: 16 km
Vehicle: Bike
Status: requested

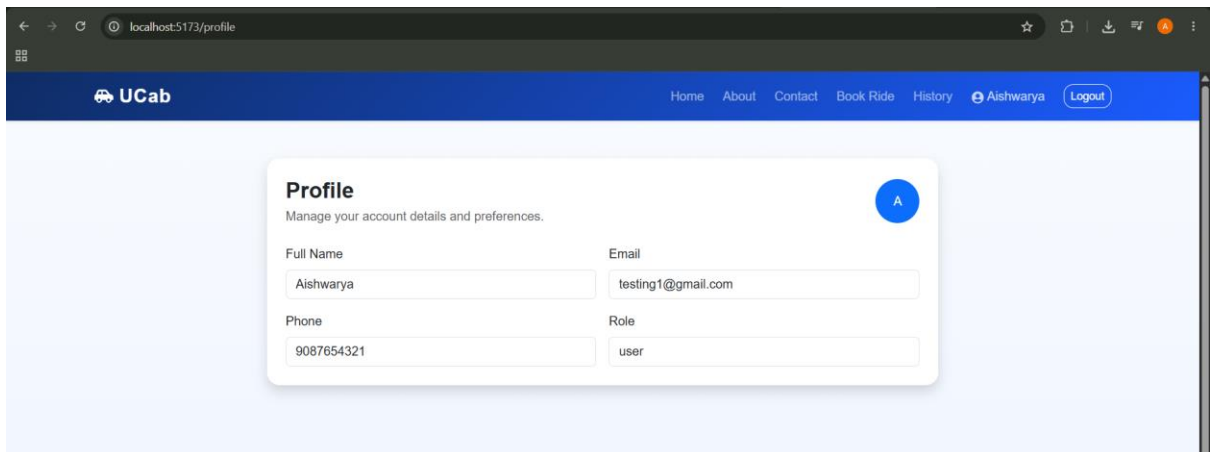
Pay now

- Booking History



Pickup	Destination	Distance	Fare	Status	Payment	Action
Ameerpet metro station	Bus station, Mehdiapatnam	16 km	₹227	requested	paid	Completed
Ameerpet metro station	Bus station, Mehdiapatnam	16 km	₹227	requested	paid	Completed
Tank bund	charminar	18 km	₹320	requested	paid	Completed
cyber towers, Hitech city	JNTU	15 km	₹275	requested	paid	Completed
railway station, rajamundry	bus stand, ravulapalem	10 km	₹200	requested	paid	Completed
railway station, Rajamundry	amalapuram bus stand	6 km	₹140	requested	paid	Completed
SIET college, cheyyeru	Bus station, Ravulapalem	5 km	₹125	requested	paid	Completed
South india shopping mall, Amalapuram	Bus station, Ravulapalem	20 km	₹350	requested	paid	Completed
SIET college, Cheyyeru	Bus station, Ravulapalem	20 km	₹350	requested	paid	Completed

- Profile Page



Profile
Manage your account details and preferences.

Full Name: Aishwarya

Email: testing1@gmail.com

Phone: 9087654321

Role: user

Google drive link:

<https://drive.google.com/file/d/16cJgrdae5HwA6eWYvNgBsdthCW1lwNbM/view?usp=sharing>

12. Known Issues

The current version of UCab has the following limitations:

- Real-time GPS tracking is simulated.
- Online payment gateway is not integrated.
- Driver and Admin dashboards are basic and can be enhanced.
- Push notifications are not implemented.
- Live driver availability is not supported.

These limitations do not affect the core booking functionality of the application.

13. Future Enhancements

The following features can be implemented in future versions:

- Live GPS Tracking
- Google Maps Integration
- Online Payment Gateway
- Driver Dashboard
- Admin Dashboard
- Push Notifications
- Ride Rating and Reviews
- Emergency SOS Feature
- Coupon and Discount System
- Favorite Locations
- AI-based Fare Prediction
- Real-Time Driver Matching
- Multi-language Support

Conclusion

UCab successfully demonstrates the implementation of a complete Full Stack MERN web application for cab booking. The project integrates frontend development, backend services, database management, RESTful APIs, authentication, and responsive UI design into a single application. It provides users with an efficient platform to book rides while showcasing practical knowledge of modern web development technologies. The project serves as a strong foundation for building scalable transportation and ride-booking systems.